

Leetcode

Functional requirements :

- user should be able to solve any problem based on his preference,
- user should be able to search for any problem based on his preference,
- user should be able to participate in coding contest based on his preference,

Non-functional requirements:

- Data-partitioning,
- load balancing,
- fault tolearance,
- vertical scalability,
- horizontal scalability,
- low latency,
- high throughput,

CAP theorem (Here we need to prioritise consistency for rank predictions during contests & for search Scenario we need to prioritise 99.99% uptime , Avbailability>>>Consistency)

Core Entities:

- user ,
- problem,
- solution,
- contest

API:

POST:/submit

```
{
  userID,
  solutionMetadata
}
```

GET:/search/{problemID}

HLD:

